

# Disk Usage Monitoring and Email Notification Using AWS SNS

## Description

1. Created a script on the EC2 instance that checked the disk usage and wrote the output to a file.
2. Verified the recipient email address through an AWS SNS email subscription.
3. Configured Amazon Simple Notification Service (SNS) to deliver notifications through email.
4. Verified that the disk usage report was successfully sent to the recipient email address.

## 1. Checked the EC2 Instance

The EC2 instance was running **Red Hat Enterprise Linux 10.2 (Coughlan)**.

The AWS CLI was checked and was initially not installed on the instance. AWS CLI was then installed using the RHEL package manager.

```
sudo dnf install -y awscli
```

The AWS CLI installation was verified using:

```
aws --version
```

## 2. Configured IAM Role for EC2

The EC2 instance initially did not have an IAM role, so the AWS CLI could not authenticate with AWS services.

An IAM role named:

```
sns-role
```

was created and attached to the EC2 instance.

The role provided the required permission for the EC2 instance to publish messages to Amazon SNS.

The role was verified using:

```
aws sts get-caller-identity
```

The command confirmed that the EC2 instance was using the `sns-role`.

### 3. Created an SNS Topic

Amazon Simple Notification Service (SNS) was used as the notification service.

An SNS topic named:

```
mytopic
```

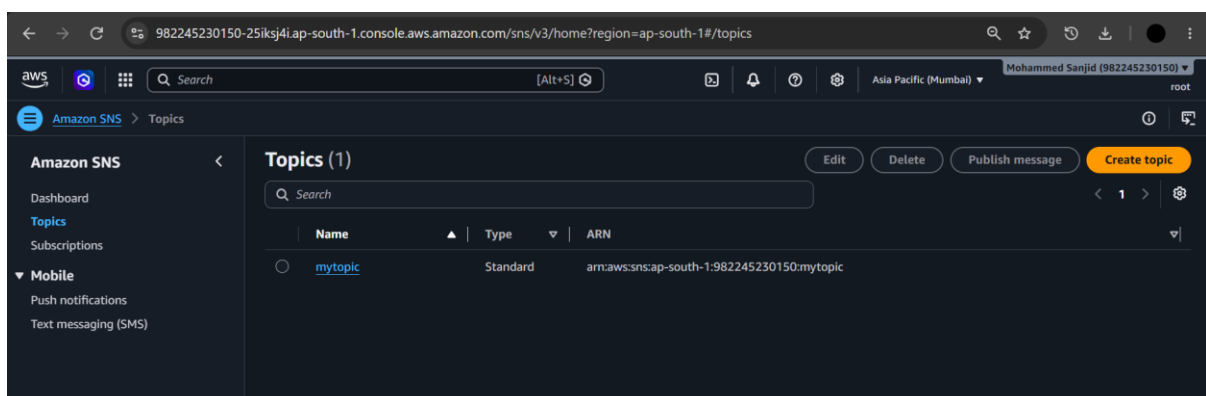
was created in the **ap-south-1** AWS Region.

The topic ARN was:

```
arn:aws:sns:ap-south-1:982245230150:mytopic
```

The SNS topic acted as the notification channel between the EC2 instance and the recipient email address.

### Output / Screenshot



## 4. Added and Verified Email Subscription

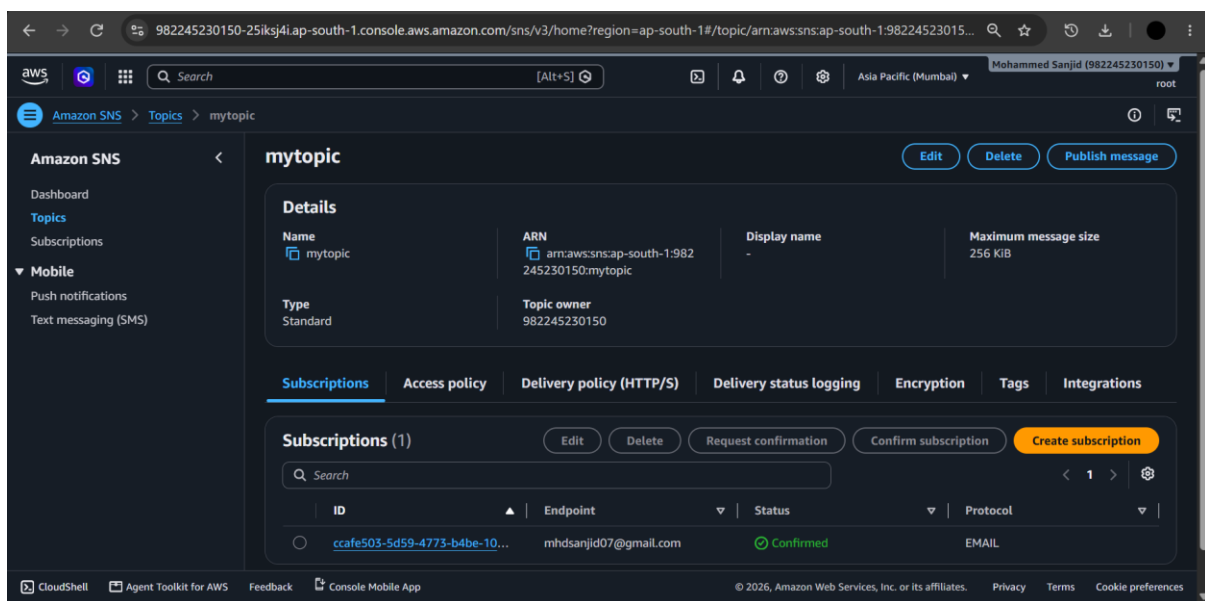
An email subscription was added to the mytopic SNS topic.

The recipient email address received an SNS subscription confirmation email.

The subscription confirmation link was opened, and the email subscription was successfully verified.

The subscription status changed from **Pending confirmation** to **Confirmed**.

### Output / Screenshot



## 5. Tested SNS Email Notification

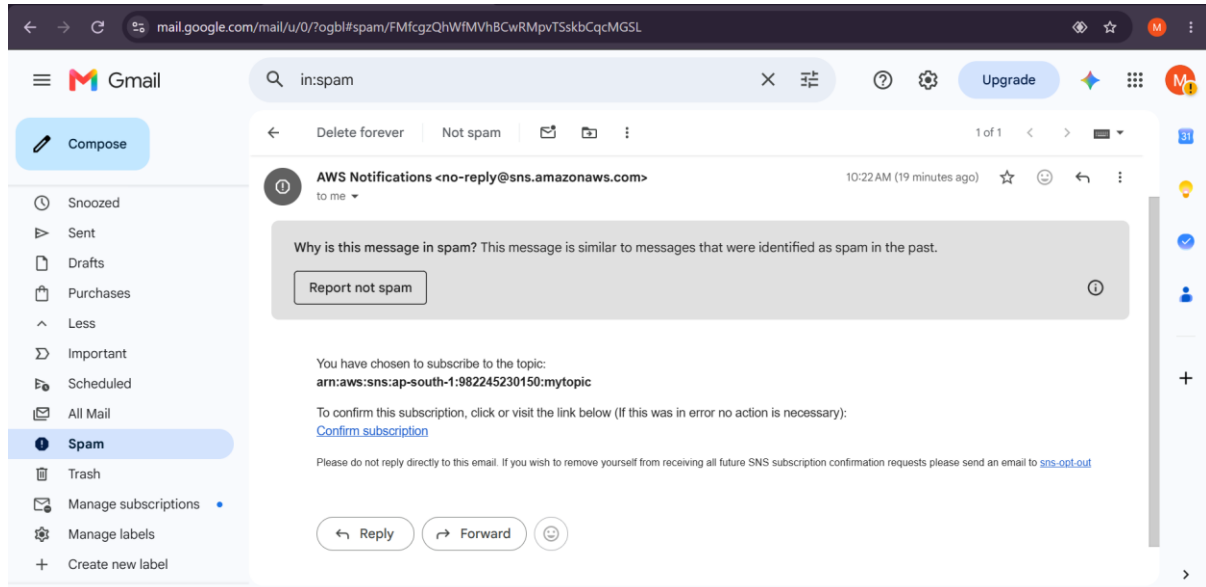
Before connecting the disk-usage script, SNS email delivery was tested using the AWS CLI.

The following command was used:

```
aws sns publish \  
  --topic-arn "arn:aws:sns:ap-south-1:982245230150:mytopic" \  
  --subject "SNS Disk Usage Test" \  
  --message "This is a test notification from the EC2 instance."
```

The message was successfully published to the SNS topic and received in the verified recipient email address.

## Output / Screenshot



## 6. Created the Disk Usage Script

A shell script named:

`disk_usage.sh`

was created on the EC2 instance.

The script initially checked the disk usage using the `df -h` command and stored the result in:

`disk_usage.txt`

The script was made executable using:

`chmod +x disk_usage.sh`

The script was executed using:

`./disk_usage.sh`

The generated report was verified using:

```
cat disk_usage.txt
```

The file contained the disk usage information of the EC2 instance.

## Output / Screenshot

```
[ec2-user@ip-172-31-6-111 ~]$ cat disk_usage.txt
Filesystem      Size  Used Avail Use% Mounted on
/dev/nvme0n1p3  9.8G  5.1G  4.7G  53% /
devtmpfs         347M    0  347M   0% /dev
tmpfs            454M    0  454M   0% /dev/shm
efivarfs        128K   3.3K  120K   3% /sys/firmware/efi/efivars
tmpfs           182M   5.3M  177M   3% /run
tmpfs            1.0M    0   1.0M   0% /run/credentials/systemd-journald.service
/dev/nvme0n1p2  200M   9.0M  191M   5% /boot/efi
tmpfs            1.0M    0   1.0M   0% /run/credentials/getty@tty1.service
tmpfs            1.0M    0   1.0M   0% /run/credentials/serial-getty@ttyS0.service
tmpfs           91M   4.0K   91M   1% /run/user/1000
```

## 7. Integrated SNS with the Disk Usage Script

The script was then modified to send the disk usage information to the SNS topic after generating the report.

The final script was:

```
#!/bin/bash
```

```
TOPIC_ARN="arn:aws:sns:ap-south-1:982245230150:mytopic"
```

```
REPORT_FILE="disk_usage.txt"
```

```
df -h > "$REPORT_FILE"
```

```
MESSAGE=$(cat "$REPORT_FILE")
```

```
aws sns publish \
  --topic-arn "$TOPIC_ARN" \
  --subject "EC2 Disk Usage Report" \
  --message "$MESSAGE"
```

The script performed the following operations:

1. Checked disk usage using `df -h`.
2. Stored the output in `disk_usage.txt`.

3. Read the contents of the file.
4. Published the disk usage information to the SNS topic.
5. SNS delivered the notification to the verified email address.

## 8. Verified the Disk Usage Email

The completed script was executed using:

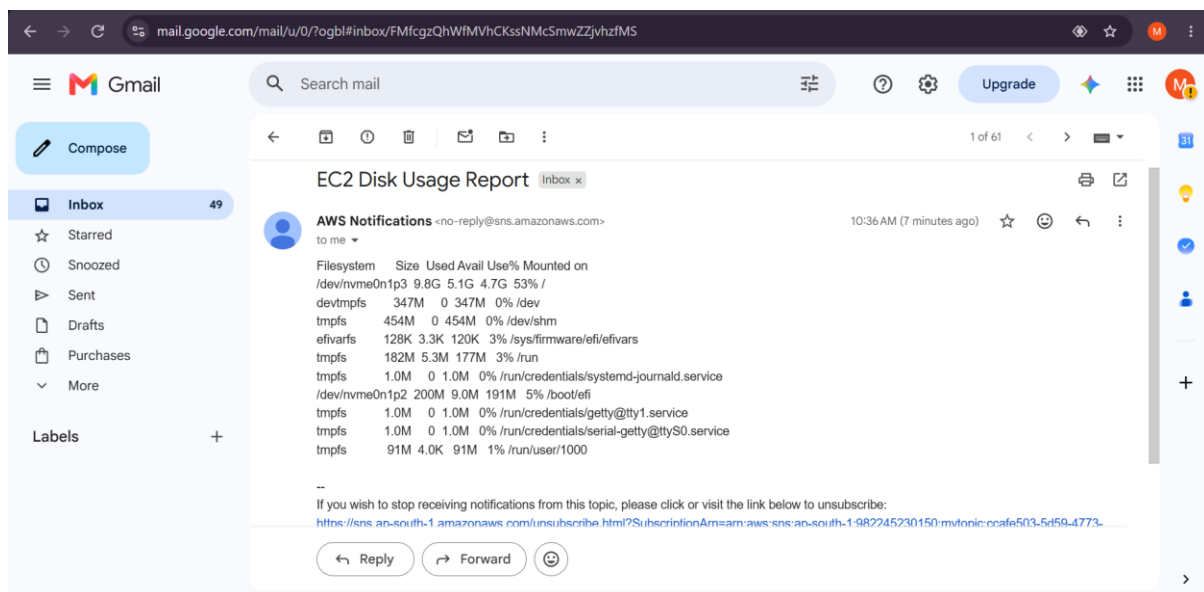
```
./disk_usage.sh
```

The command successfully published the disk usage information to the SNS topic.

The recipient email received the **EC2 Disk Usage Report**, containing the output generated by `df -h`.

This confirmed that the complete notification process was working successfully.

### Output / Screenshot



## Conclusion

The disk usage monitoring and email notification system was successfully configured on the RHEL EC2 instance. A shell script was created to collect disk usage information and save it to

a file. Amazon SNS was configured with a verified email subscription, and the EC2 instance was given the required IAM permissions to publish notifications. The script successfully sent the disk usage report through SNS, and the report was received in the verified email address.